FrutaMIDI: capacitive USB MIDI controller

Engineering report

FrutaMIDI engineering project

Revision: `TO BE FILLED FROM EVIDENCE`

Abstract

TO BE FILLED FROM EVIDENCE. One-paragraph summary of scope, verification status and open risks, written last from the gate summaries.

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1 Introduction and scope

FrutaMIDI is a USB MIDI controller that turns touches on conductive objects (fruit, plants, water) into MIDI notes, using a Raspberry Pi Pico and an MPR121 capacitive sensor. It is built either on a breadboard or on a single-sided PCB in `hardware/`.

1.1 Scope

TO BE FILLED FROM EVIDENCE. In-scope and out-of-scope items, from management/PROJECT_DEFINITION.md.

1.2 Document conventions

- Every number carries one label: SOURCE, CALCULATION, SIMULATION, MEASUREMENT or ASSUMPTION.
- Status is reported only as PASS, PARTIAL, FAIL, BLOCKED or NOT RUN.
- Passing ERC, DRC or a compile is not hardware validation.

2 Requirements and verification status

Source: `management/REQUIREMENTS.md`. The table below is generated from a CSV export into `generated/requirements_table.tex`; do not edit it by hand.

TO BE FILLED FROM EVIDENCE. Requirements table (ID, statement, verification method, status, evidence path). Generated from the requirements CSV; the file `generated/requirements_table.tex` does not exist yet.

3 Architecture and decisions

3.1 System overview

TO BE FILLED FROM EVIDENCE. Block diagram and description: electrodes, MPR121, Pico, USB MIDI, host sampler/DAW. Figure from reports/figures/.

3.2 Decisions

TO BE FILLED FROM EVIDENCE. Summary of each ADR in management/DECISIONS.md: context, decision, consequences, status.

4 Interfaces

Source: `management/INTERFACES.json`.

TO BE FILLED FROM EVIDENCE. Electrical interfaces (pins, voltages, I^2C addresses), USB MIDI messages, and host interfaces. Each value labelled.

5 Electronics analysis

5.1 Schematic and PCB

TO BE FILLED FROM EVIDENCE. Design summary, ERC/DRC results with report paths and revision. ERC/DRC passing is not hardware validation.

5.2 Analyses

TO BE FILLED FROM EVIDENCE. Power budget, sensing range, ESD and current limiting on touchable electrodes (T-050), with assumptions, sensitivity and, where applicable, convergence.

5.3 Safety statement

TO BE FILLED FROM EVIDENCE. Final wording of the safety statement (ADR-003), based on the analysis above.

6 Firmware design and tests

Source: `firmware/FrutaMIDI/`.

6.1 Design

TO BE FILLED FROM EVIDENCE. Module structure, main loop, touch/release handling, scales, octave, aftertouch, recalibration.

6.2 Build and tests

TO BE FILLED FROM EVIDENCE. Compile results per board (tool version, result line), host unit tests (command, pass count), with log paths and revision.

7 Host software and tests

Source: `docs/index.html` (browser sampler).

TO BE FILLED FROM EVIDENCE. Sampler design, MIDI handling, tests performed (command, result line), browser compatibility evidence.

8 Independent review findings

TO BE FILLED FROM EVIDENCE. Findings from the independent reviewer: ID, severity, description, disposition (fixed / accepted / open), evidence path.

9 Limitations

TO BE FILLED FROM EVIDENCE. Known limitations, untraced claims, assumptions not yet checked and open risks from management/RISKS.md.

10 Physical validation plan

No part of the hardware has been measured. Each row below defines a physical test still to run. Pass criteria come from the requirements, not from this report.

What to measure	How (instrument, setup)	Pass criterion	Status
TO BE FILLED FROM EVIDENCE			NOT RUN

11 Reproducibility

11.1 Toolchain

TO BE FILLED FROM EVIDENCE. Tool versions from management/TOOLCHAIN.md (KiCad, arduino-cli and cores, Python, Node, TinyTeX and installed packages).

11.2 How to rebuild

- Hardware: `sh hardware/scripts/build_all.sh` (regenerates, runs ERC/DRC, exports fabrication files).
- Report: MCP `eng-tools latex_build` with `reports/report/frutamidi_report.tex`.

TO BE FILLED FROM EVIDENCE. Firmware compile and test commands, released file checksums (releases/V1_BASELINE/).

A Sources

Source list: `management/SOURCES.md`.

TO BE FILLED FROM EVIDENCE. Datasheets, standards and external references cited in this report, with identifier and access date.